Time Series Analysis

PARAMI UNIVERSITY

Time Series Analysis

Today's Topic: Autoregressive Integrated Moving Average Model

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https://sithuaung-ink.github.io/data-analyst-portfolio/

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Properties of ARMA Model

- The ARMA(p, q) is stationary if and only if its AR part is stationary.
- The ARMA(p, q) is invertible if and only if its MA part is invertible.
- The ARMA(p, q) is causal if and only if its AR part is causal.

Properties of ARMA Model

The definition of ARMA models is as follows.

$$y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-p}$$

- The AR part controls dependence on past values.
- The MA part controls dependence on past errors.

Properties of ARMA Model

Example: Given AR(1)

$$y_t = \varphi_1 y_{t-1} + \epsilon_t$$

If:

$|\varphi| < 1$, stationary.

$|\varphi| \geq 1$, not stationary.

Even if we add MA terms:

$$y_t = \varphi_1 y_{t-1} + \epsilon_t + 0.5\epsilon_{t-1}$$

If:

$|\varphi| < 1$, time series is stationary.

$|\varphi| \geq 1$, time series is not stationary.

Properties of ARMA Model

Example: Given MA(1)

$$y_t = \epsilon_t + \theta_1 \epsilon_{t-1}$$

If:

$|\theta| < 1$, invertible.

$|\theta| \geq 1$, not invertible.

Properties of ARMA Model

Example: Given AR(1)

$$y_t = \varphi_1 y_{t-1} + \epsilon_t$$

If:

$|\varphi| < 1$, causal

$|\varphi| \geq 1$, non causal.

Autoregressive Moving Average Model

The definition of ARMA models as follows.

$$y_t = \varphi_1 y_{t-1} + \varphi_2 y_{t-2} + \ldots + \varphi_p y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-p}$$

ARMA combines both past values and past errors.

Autoregressive Integrated Moving Average Model

ARIMA (p, d, q)

p = AR order

q = MA order

d = differencing order

Normality Test

normaltest (From Scipy Package)

Null hypothesis:

H_0 : Residuals are normally distributed

Alternative

H_1: Residuals are not normally distributed

Interpretation:

p-value < 0.05 -> reject H_0 -> Residuals are not normally distributed

p-value > 0.05 -> fail to reject -> Residuals are normally distributed

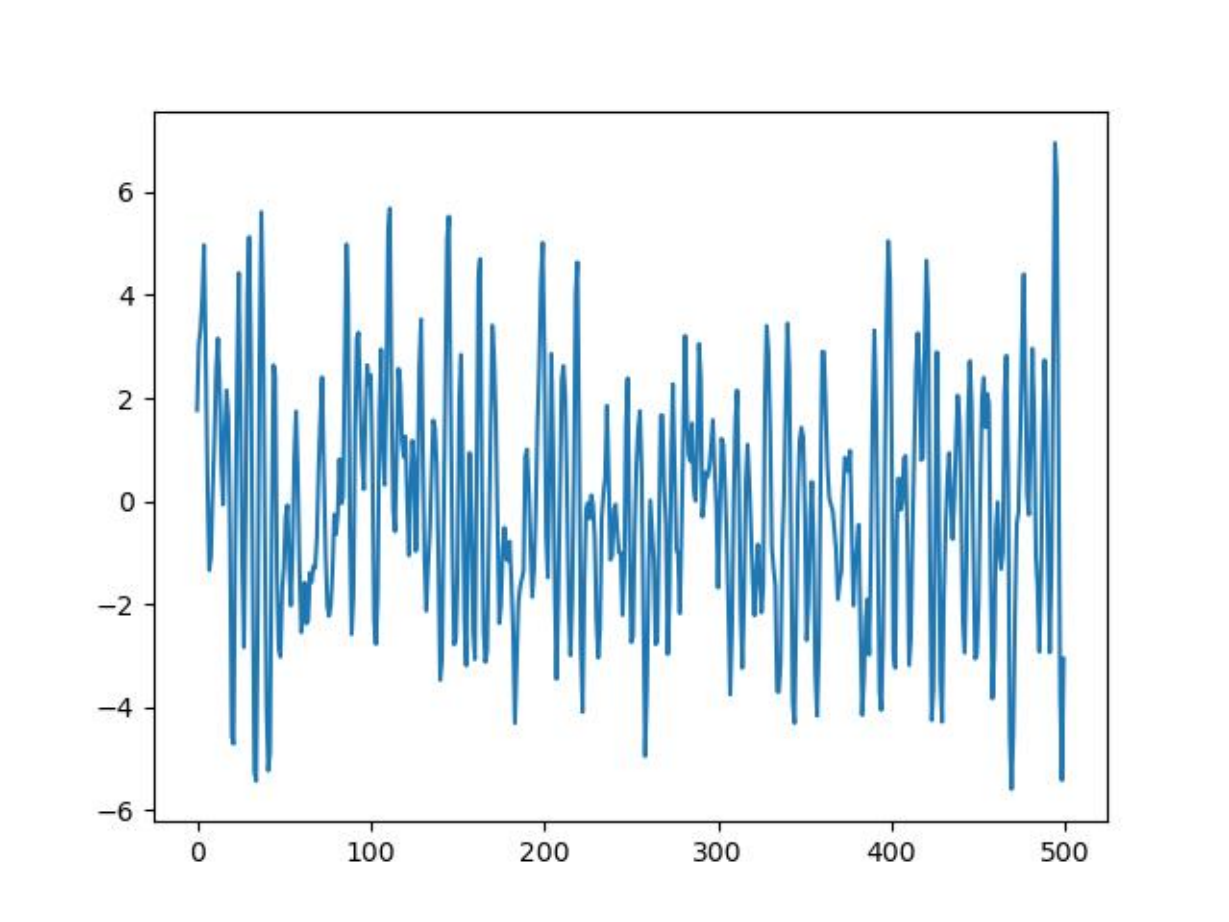
ARIMA(2,0,2)

$$y_t = 0.8y_{t-1} - 0.6y_{t-2} + \epsilon_t + 0.7\epsilon_{t-1} + 0.4\epsilon_{t-2}$$

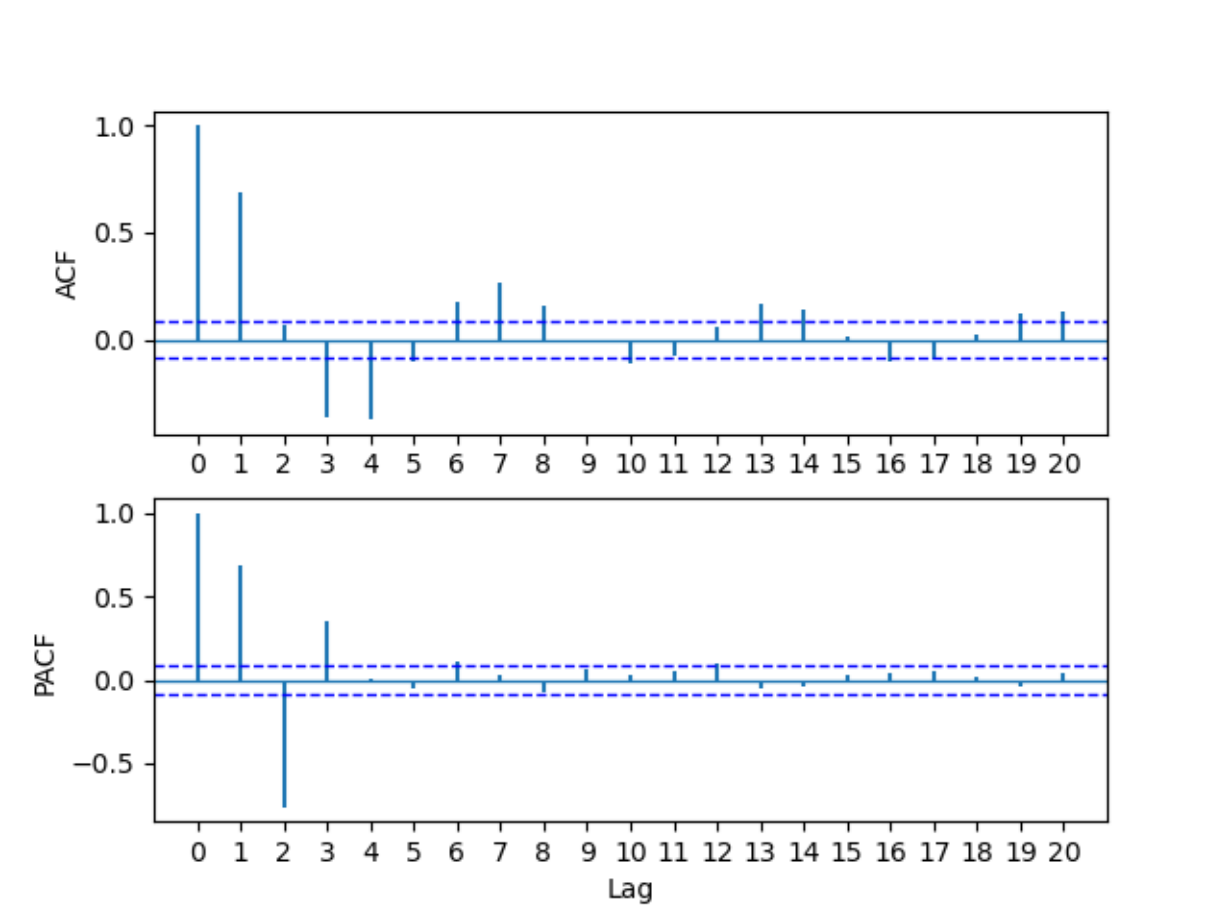
$$y_t - 0.8y_{t-1} + 0.6y_{t-2} = \epsilon_t + 0.7\epsilon_{t-1} + 0.4\epsilon_{t-2}$$

$\varphi_1 = -0.8$, $\theta_1 = 0.6$ $\varphi_2 = 0.7$, $\theta_2 = 0.4$

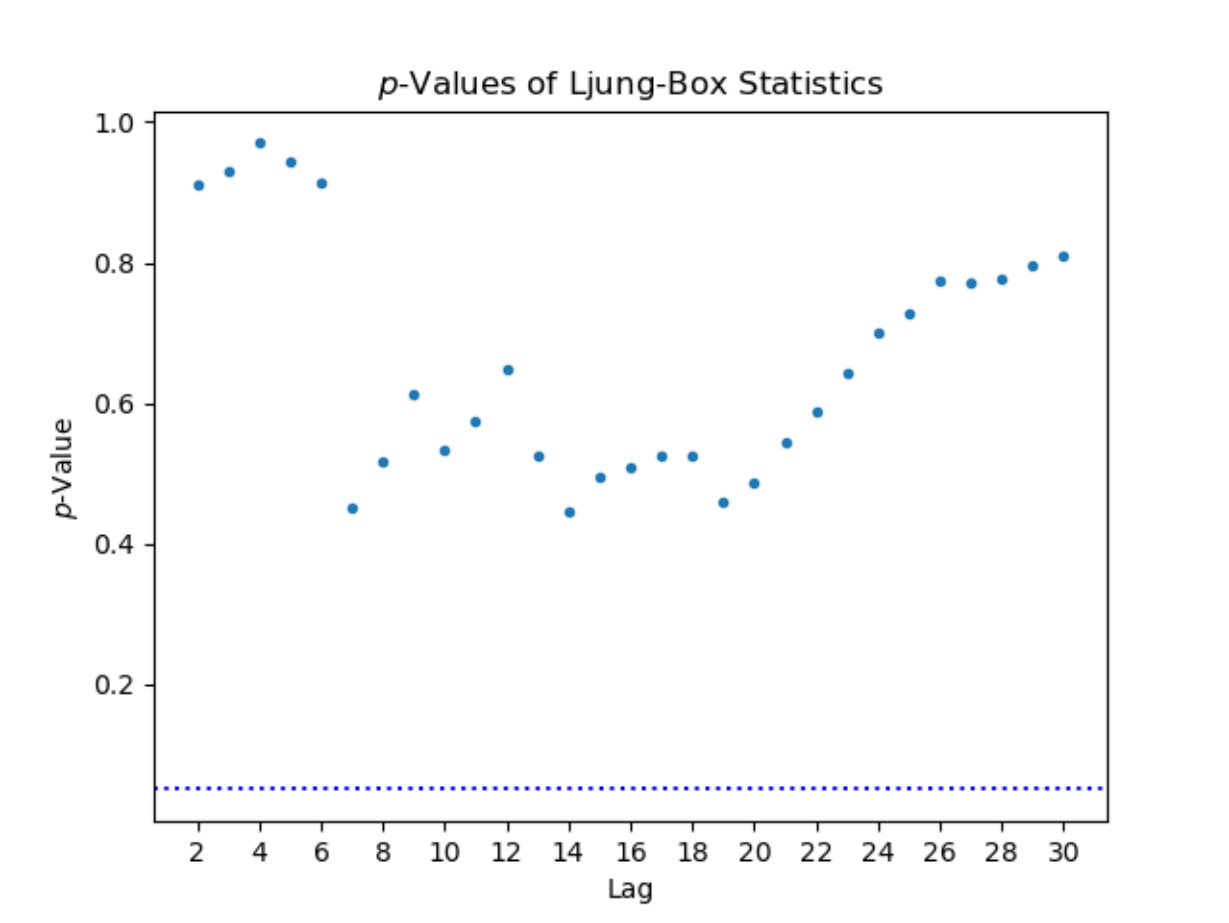
ARIMA(2,0,2)



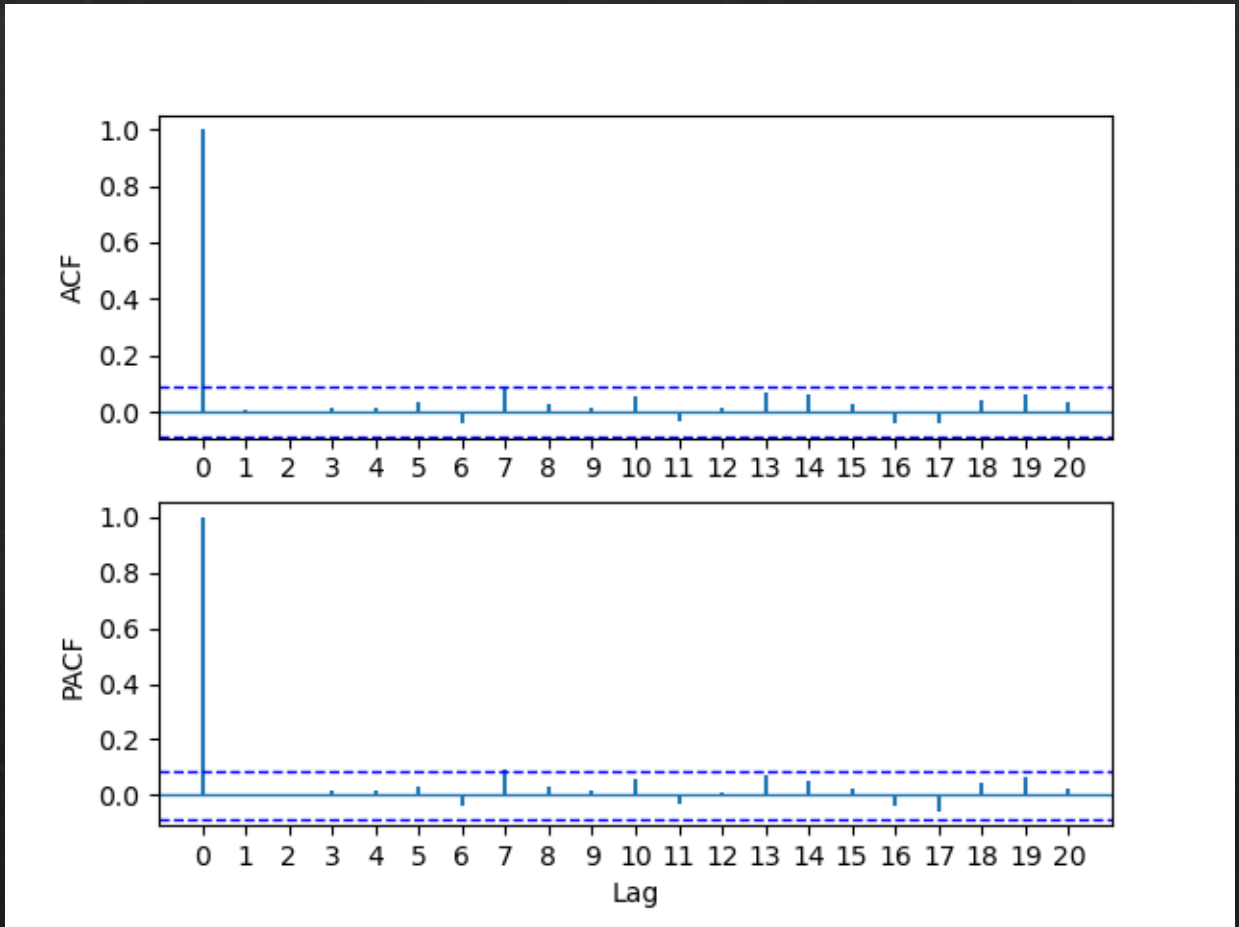
ARIMA(2,0,2)


ACF
1.0
0.5
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
PACF
1.0
0.5
0.0
−0.5
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Lag

ARIMA(2,0,2)


p-Values of Ljung-Box Statistics
1.0
0.8
0.6
0.4
0.2
p-Value
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
Lag

ARIMA(2,0,2)


ACF
1.0
0.8
0.6
0.4
0.2
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
PACF
1.0
0.8
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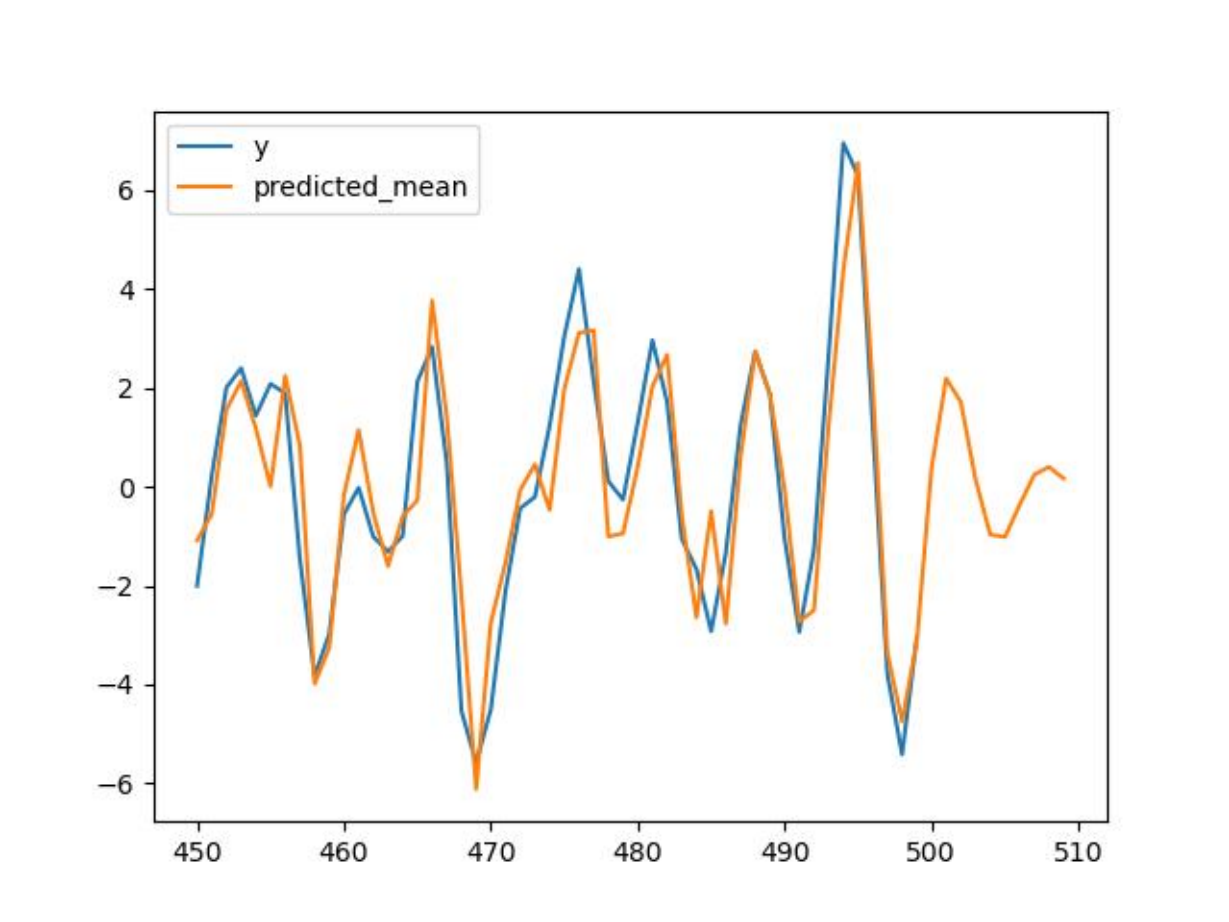
0.0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Lag

ARIMA(2,0,2)



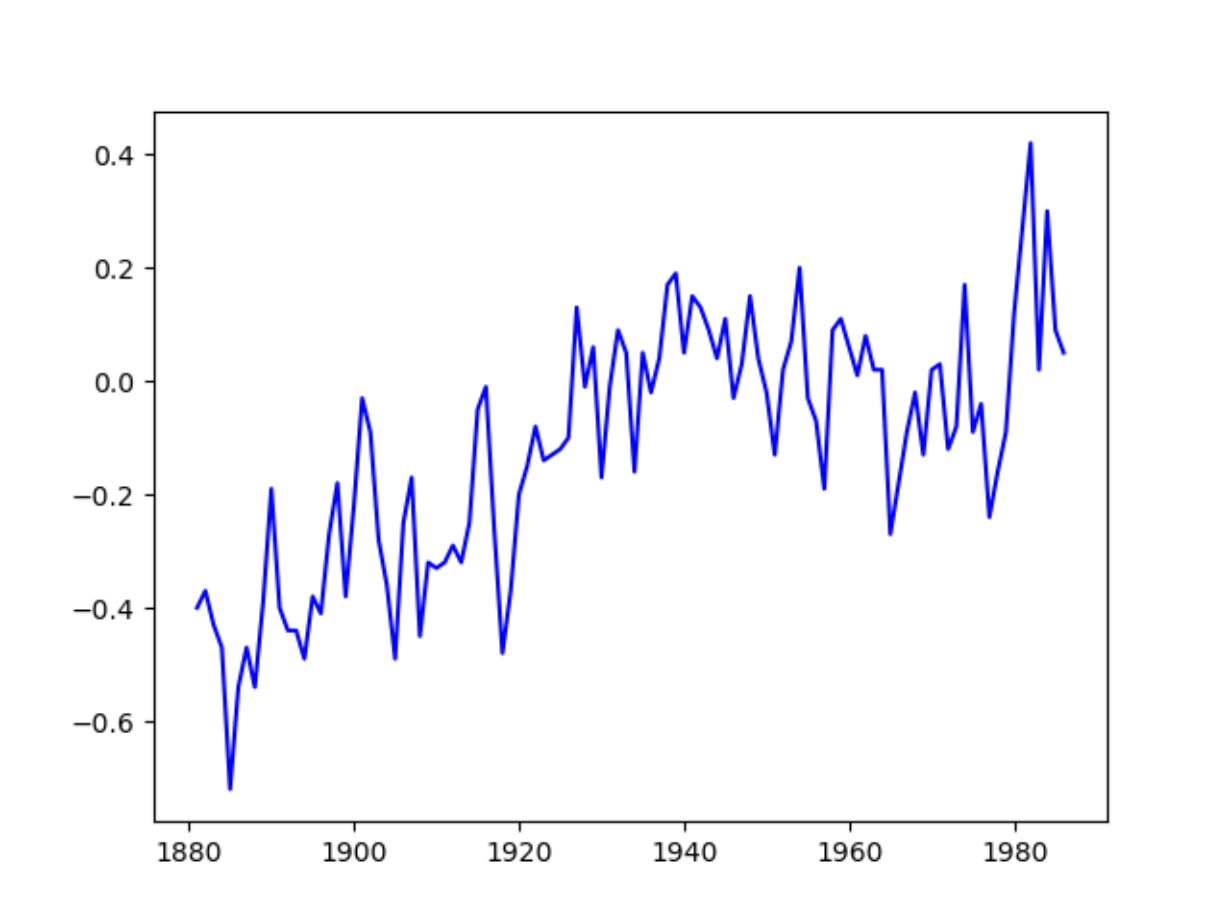
Global Annual Mean Surface Air Temperature

Dataset description

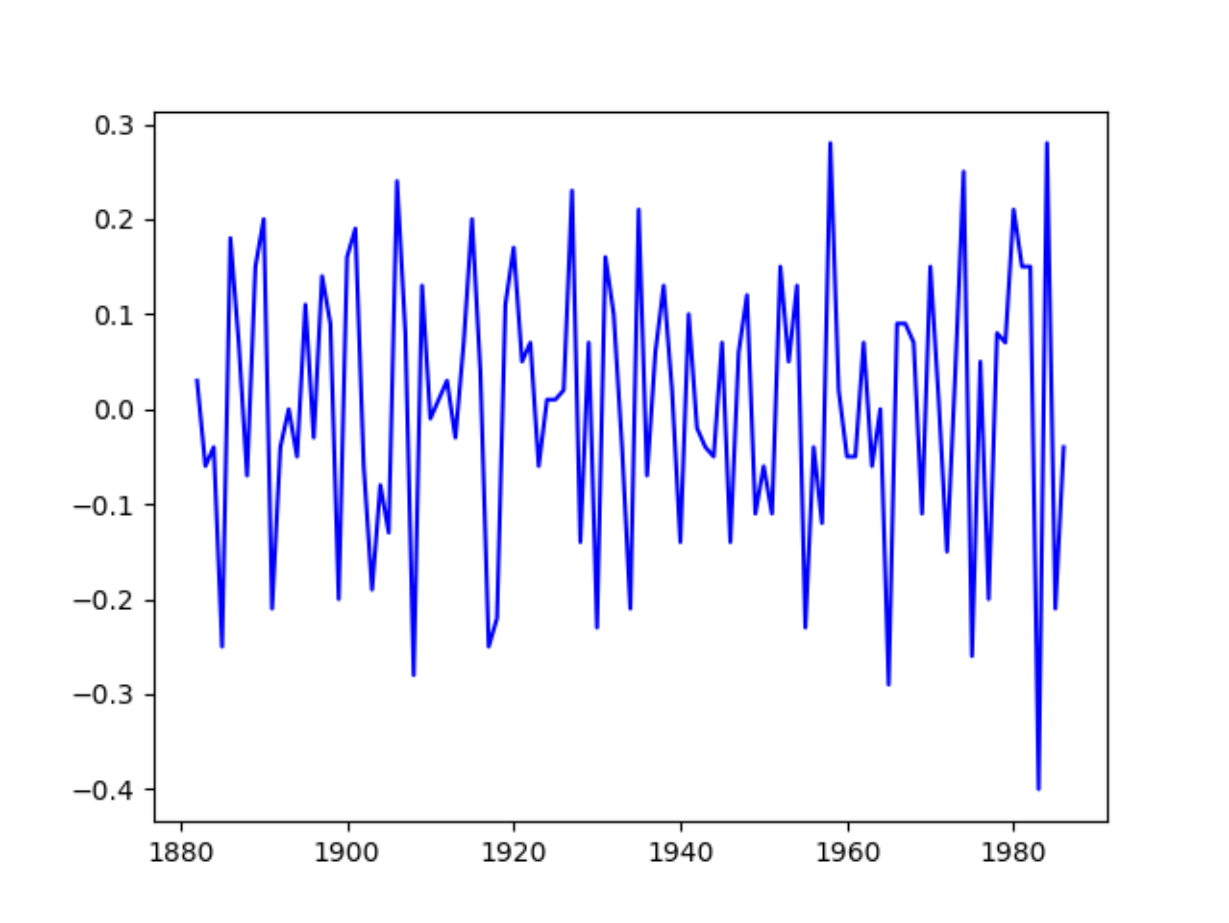
- The time series dataset "Global mean surface air temp changes 1880 to 1985) is from Hansen and Lebedeff (1987)[1].
- The dataset, which is an investigation of the global warming issue.

[1]Hansen, J. and Lebedeff, S. (1987) Global Trends of Measured Surface Air Temperature. Journal of Geophysical Research, 92, 13345-13372. http://dx.doi.org/10.1029/JD092iD11p13345

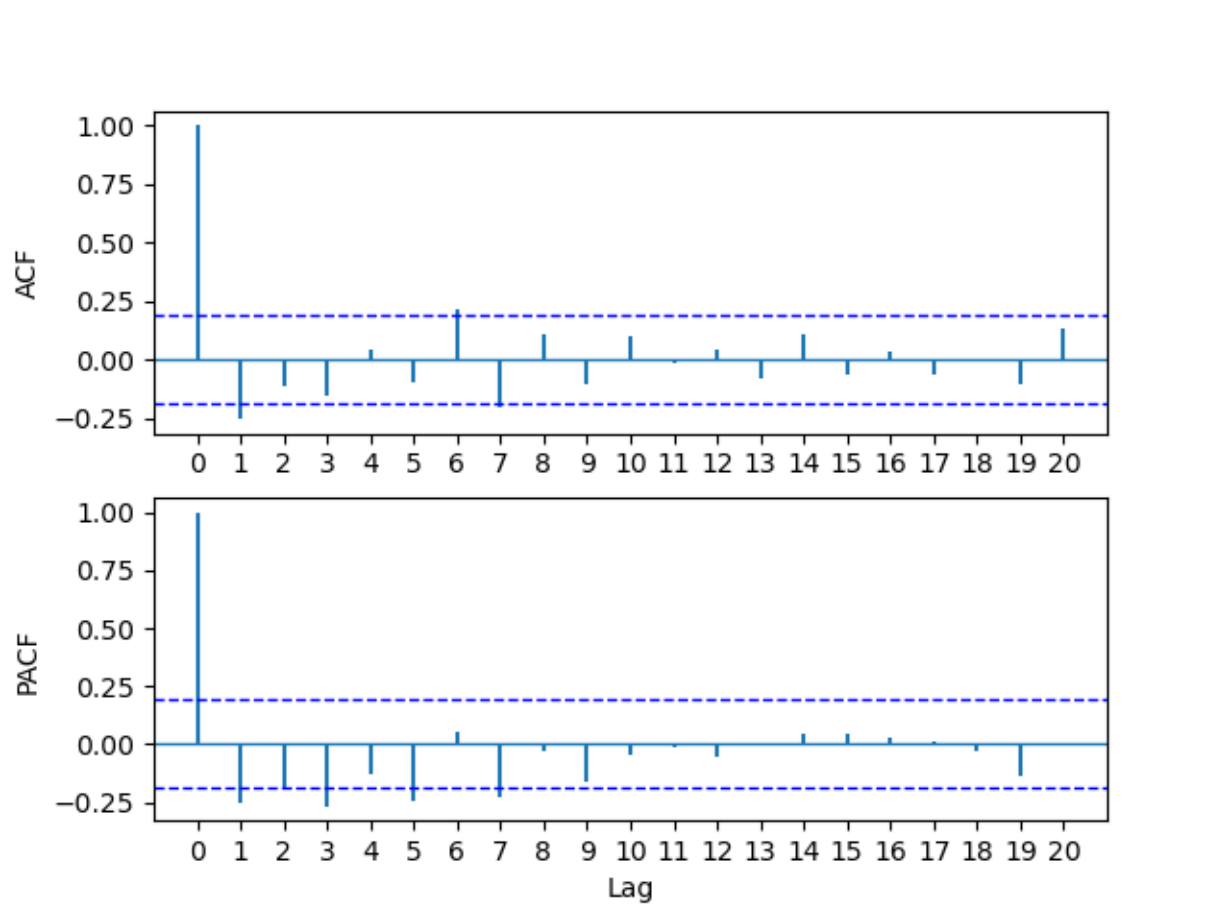
GMSATC



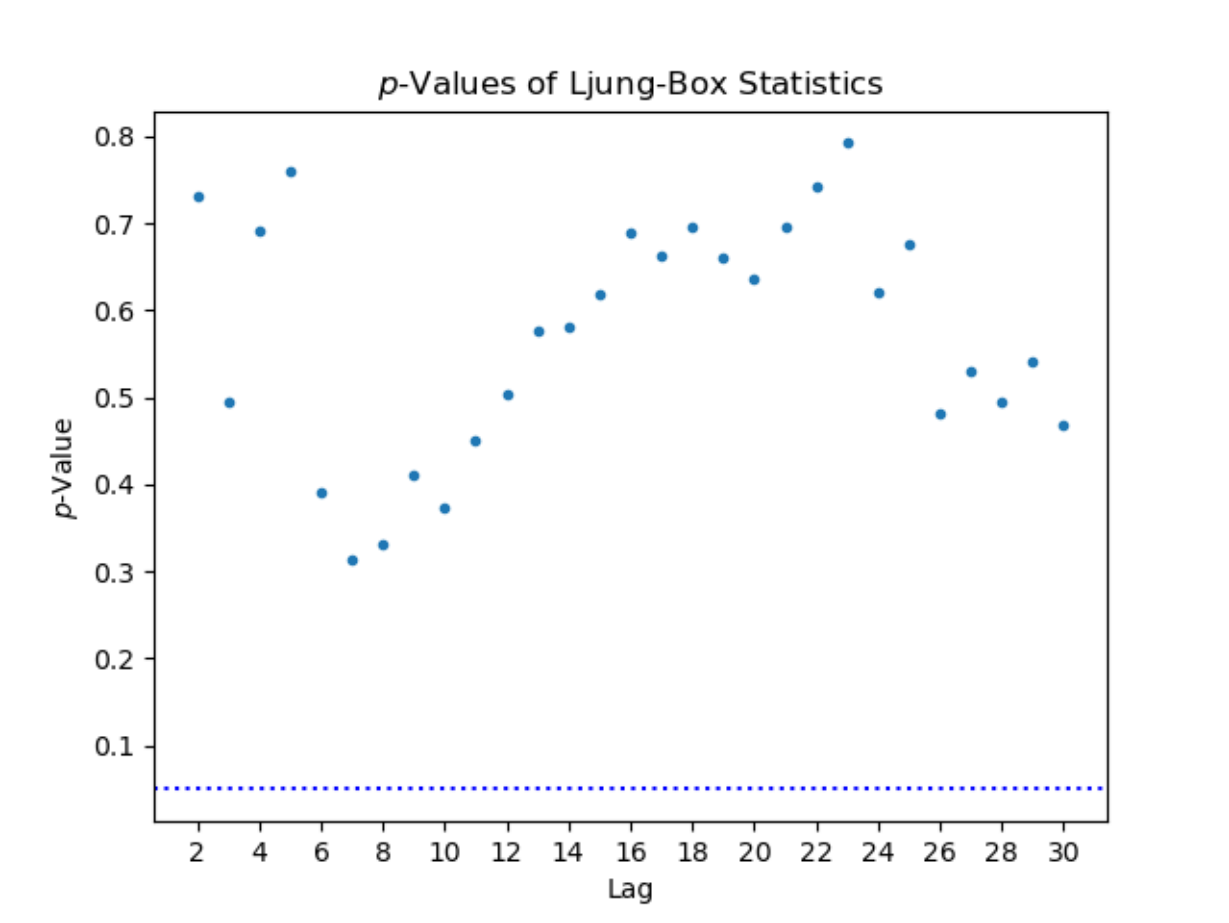
GMSATC



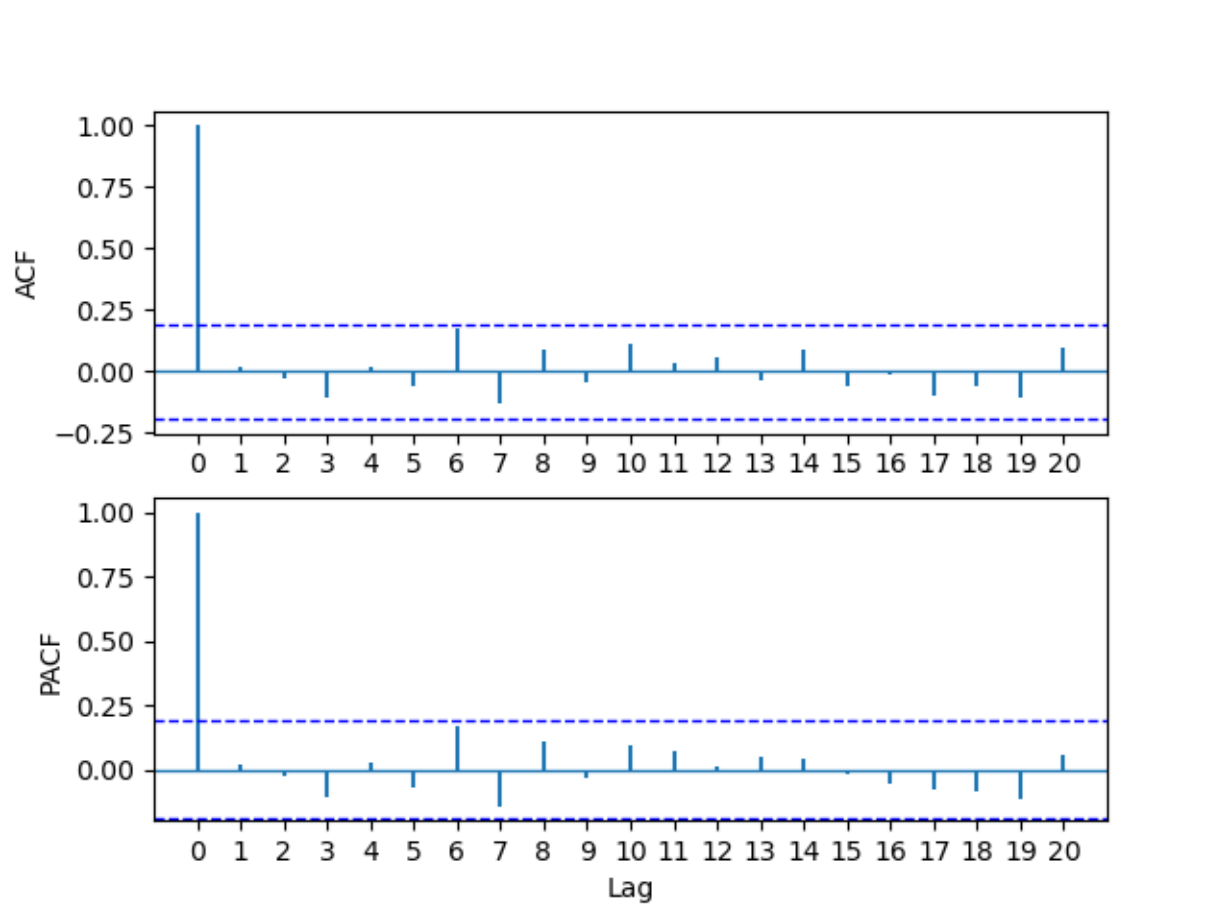
GMSATC


ACF
1.00
0.75
0.50
0.25
0.00
−0.25
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
PACF
1.00
0.75
0.50
0.25
0.00
−0.25
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Lag

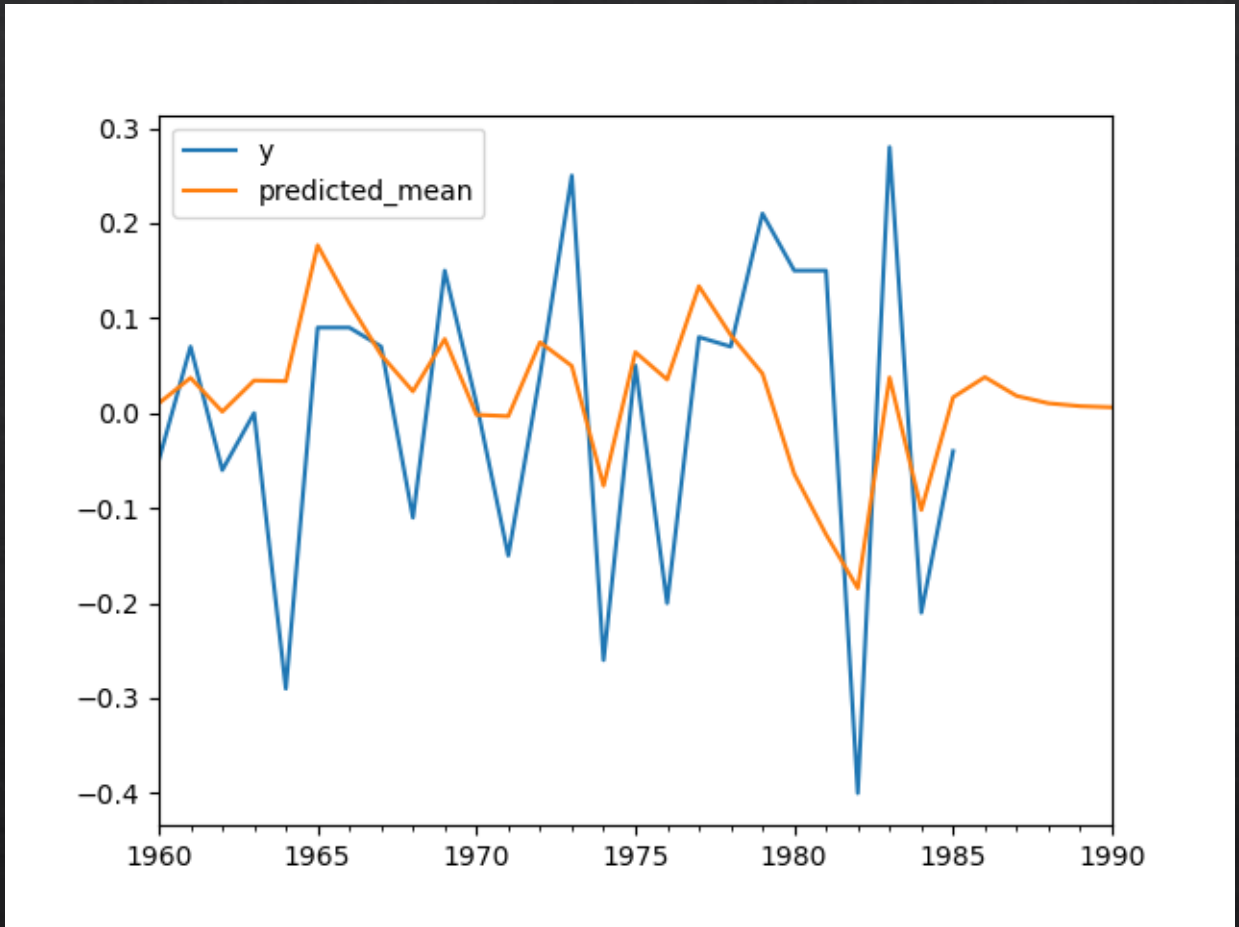
GMSATC


p-Values of Ljung-Box Statistics
p-Value
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
Lag

GMSATC


ACF
1.00
0.75
0.50
0.25
0.00
−0.25
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
PACF
1.00
0.75
0.50
0.25
0.00
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
Lag

GMSATC


y
predicted_mean
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
1960
1965
1970
1975
1980
1985
1990